Portfolio Map



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MODULE: Architecture & operation system
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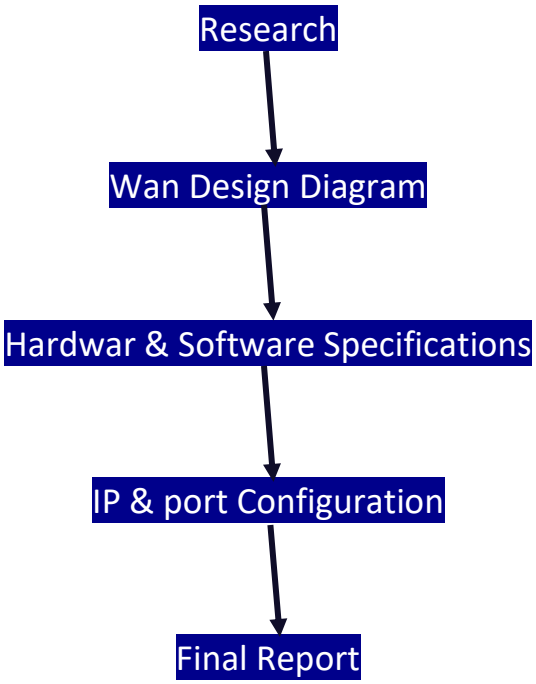
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Evolution Diagram:



Descriptions:

Research:

- The goals and specifications for a WAN architecture that would accommodate Big Data workloads were set in the first stage of this project. In modern business, a WAN must permit data transfer with low latency, providing a scalable, secure, and high-speed network.
- The study attempted to understand some of the critical issues faced in Big Data transmission, including data flow security and access speed to storage resources. Reliability and fault tolerance required emphasis on data processing and storage distribution as well as redundancy.
- During this phase, available technologies as well as network topologies and business services requirements for unstructured and structured data handling were analyzed. Primary concerns included dual cloud/on-premises support, engineered server-side communication, and a flexible IP allocation scheme. Performance had to keep pace with future data growth. This foundational work is what guided the design choices made in the later stages of the project where all strategic decisions had to be based on practical needs and flexible scaling standards for the business.

Wan Design Diagram:

- The structure of excluding big data operations is depicted in the WAN design diagram. This diagram is an illustration of the network structure that includes all aspects of the structure and relationships.
- The diagram includes PCs, Laptops, Structured Servers, Unstructured Servers, Backup servers, configuration servers, Switches, Router with firewall and Cloud Access. Each type is annotated with the data movement direction between segments.
- The network architecture promotes efficient traffic flow between multiple data zones and redundancy and scalability design features to support large data volume transfers.
- The router is connected to a firewall and to the cloud, providing limited external linking. The machines are categorized by the steps in the data processes for which they have particular applicability, for example data processing and management. The figure provides an illustration to facilitate the understanding of the system structure and helps to visualize and verify the whole system. It helps the users to understand how components interact with each other and brings clarity to the resilience and efficiency of the structure to advances of the Big Data WAN requirements.

Hardware & Software Specifications:

- The following section describes the specific hardware and software components used to deploy the WAN architecture. Each of the selected devices was determined according to performance, reliability, and compatibility with the need for Big Data processing Servers such as the HP ProLiant Gen10 and Dell OptiPlex 5070 were used with their capacity for running intensive programs.
- Switches (such as the TP-Link TL-SG1024DE) offer high-speed connectivity within network segments. The router (Mikrotik RB760iGS) and firewall (Fortinet FortiGate 30E) protect and route traffic efficiently. In terms of software, the project used a dual operating system approach: Windows Server 2019 and Ubuntu Server 22.04 for server stability and server administration tasks, and Windows 10/11 Pro for client computers.
- Central server roles were allocated such as DHCP, DNS, Active Directory, and Web/Mail services. Security software used Windows Defender for Endpoint, and Malwarebytes as an antimalware tool. Cloud-based encryption used Sync.com. With these hardware and software components combined, a safe and efficient environment with the capacity for running structured and unstructured Big Data within an efficient WAN infrastructure was provided.

IP Addressing & Port Configuration:

This section has comprehensive tables that specify the IP address and port configurations of the entire WAN network. Each device ranging from servers to client PCs and networking hardware has a specific IP address within an organized subnet (192.168.1.x).

The addresses guaranteed each device communicated efficiently with no conflicts and still left a clean and orderly structure for convenience of administration. Subnet masks were configured as 255.255.255.0 to ensure consistent boundaries of communication. Gateways were routed by the central router with an address of 192.168.1.1. Ethernet interfaces such as FastEthernet (Fa0) and GigabitEthernet (Gi0) were allocated according to the roles of the devices and the connections. Cable types such as straight-through and crossover were allocated per link to facilitate right physical communication. The port configuration also determined which switches communicated with which devices to ensure efficient paths of transmission and eliminate bottlenecks.

The configuration guarantees stable and-scalable transmission of communications among servers, clients, and outside services. Accuracy in the configuration is

paramount in guaranteeing trouble-free operation and efficient maintenance of the infrastructure of the Big Data.

Final Report:

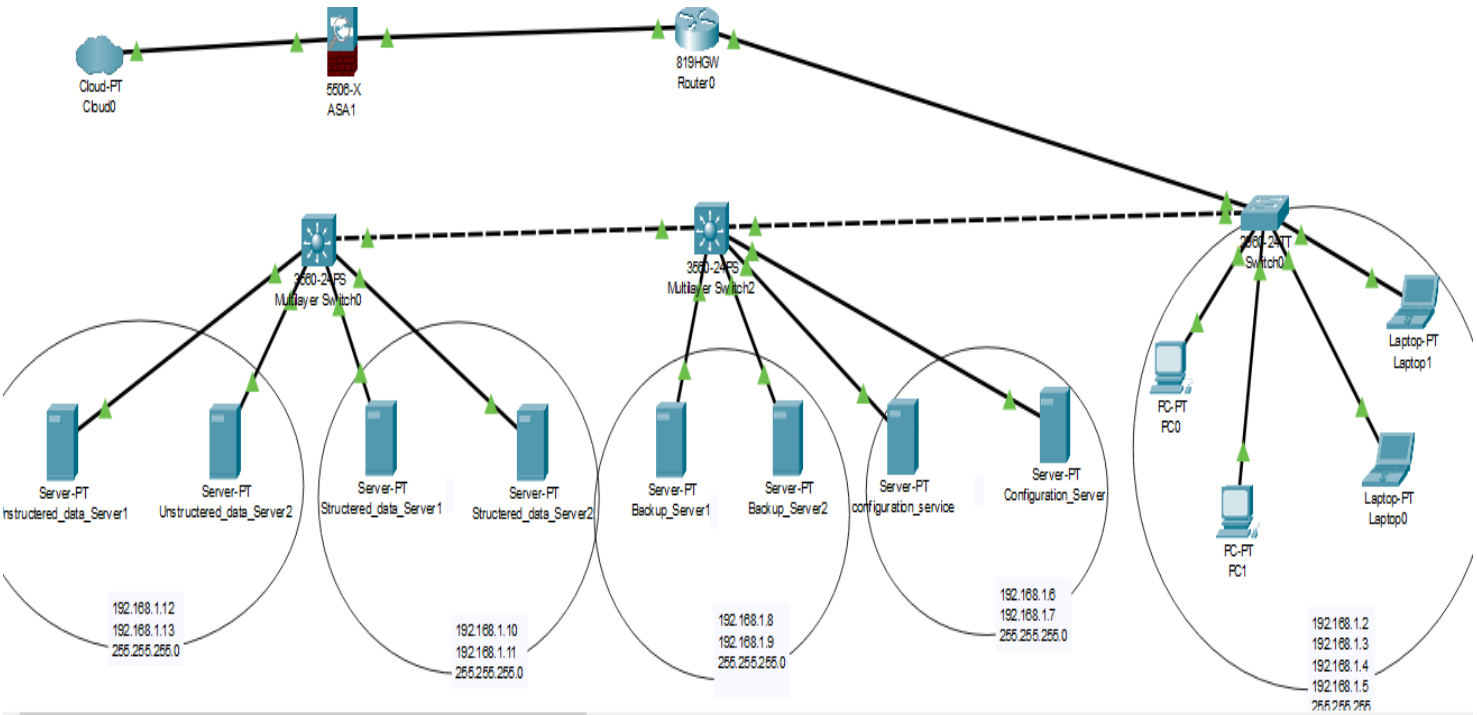
- The concluding report brings all aspects of the WAN design project into one complete, structured document. It summarizes the entire process from the initial planning stages and research up until the Implementation and configuration.
- The report breaks down how each decision in the design contributes towards the primary objective of empowering Big Data infrastructure with performance, scalability, and security. It discusses how the architecture facilitates distributed processing, centralized management, and guaranteed backup of both structured and unstructured forms of data. Redundancy is incorporated both in network paths as well as storage facilities for the prevention of a complete loss of data and ensuring uptime.
- The issue of future-proofing is also discussed with proposals for the potential scaling up of the network with growing demands of the data. The report mentions all the hardware and software used along with their sources and rationale behind the choice. By providing technical observations, configuration details and performance factors, the concluding report is a complete proof of the preparedness of the WAN architecture in handling enterprise-level environments of Big Data as well as changing technology paradigms.

Files:

Research file:

wan-architecture-and-design.pdf

Wan Design Diagram:



Hardware & software Specifications:

Hardware & Software.pdf

IP Addresses & Ports Tables:

IP Addresses & Ports.pdf 1.pdf

References

Cisco Systems, Inc. (n.d.) *WAN Architecture and Design*. [PDF] Available at: https://www.cisco.com/c/dam/en/us/solutions/collateral/enterprise-networks/intelligent-wan/wan-architecture-and-design.pdf [Accessed 13 May 2025].